

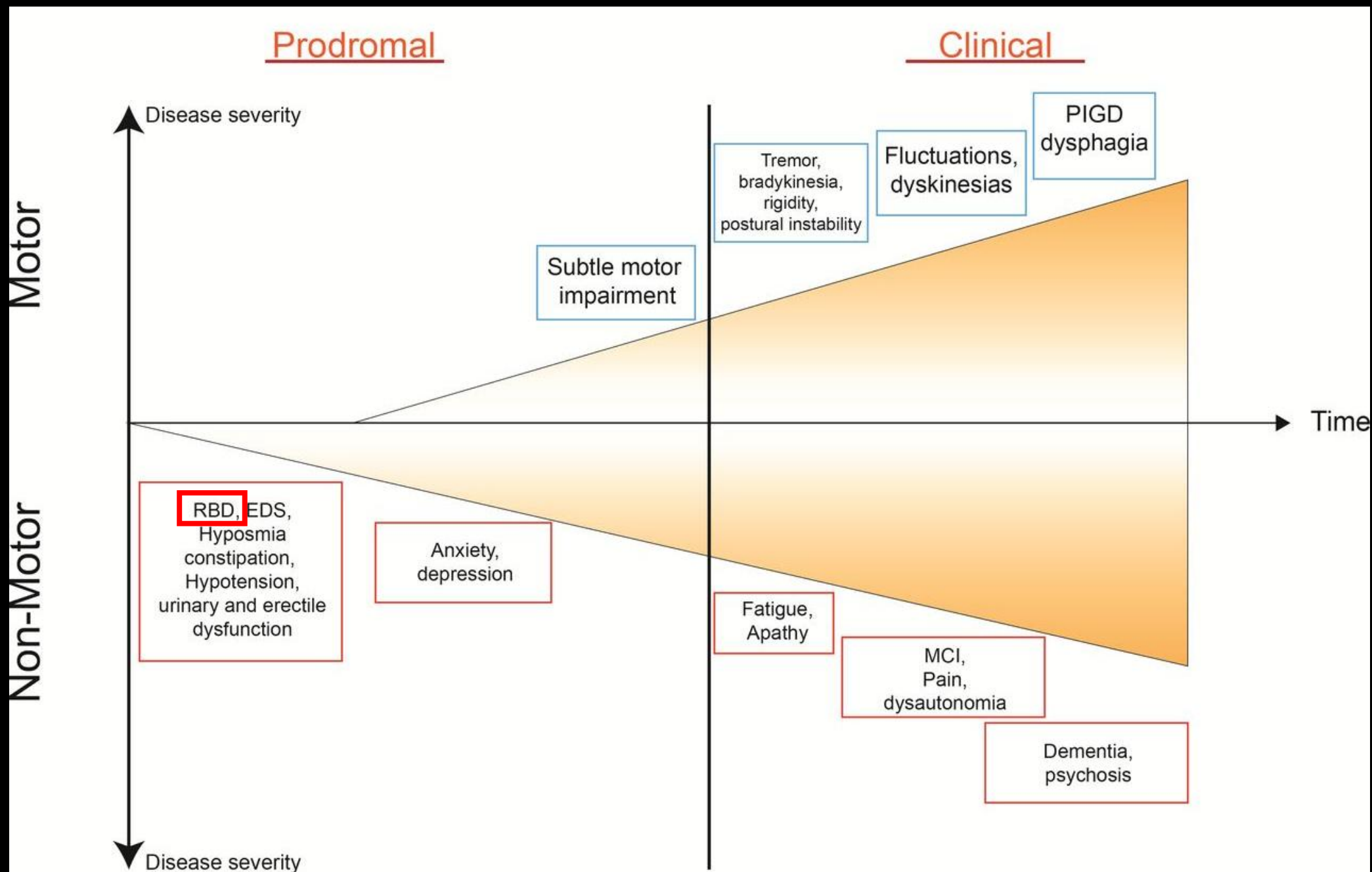
From PSG Signals to Deep Learning - An AI-Assisted Hands-on Approach -

Po-Yu Lin, MD, MSc

National Cheng Kung University Hospital, Taiwan

What this workshop session is

- A walkthrough of the most basic deep learning workflow
- Experience how AI-assisted coding changes what's possible for us
- Not a Python crash course
- Not a deep learning crash course



American Academy of Sleep Medicine

- Repeated episodes of sleep-related vocalization, complex motor behaviors, or both.
 - Repetitive episodes during a single night of video-polysomnography.
- Behaviors occur during REM sleep.
 - Shown by polysomnography, or are presumed to occur during REM sleep on the basis of clinical history of dream enactment.
- Polysomnographic recording shows REM sleep without atonia.
- The disturbance is not better explained by another sleep disorder, mental disorder, drug, or substance use.

Goal

- Lets try build a deep learning model detecting REM sleep without atonia (RSWA) in polysomnography.
 - To make things easier, the question we explore is “classification between normal REM and RSWA”.
- Let’s use it to annotate REM sleep behavior disorder in patients.

Outline

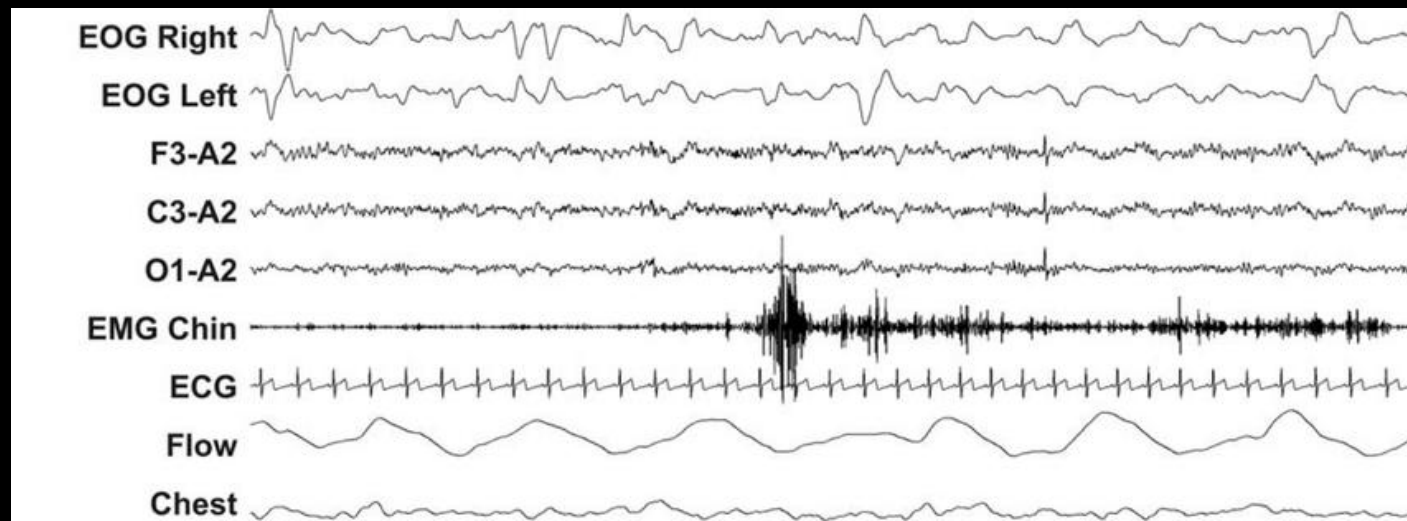
- Module 1 · From PSG signals to training data
- Module 2 · Teaching a 1D-CNN to detect RSWA
- Module 3 · Fighting overfitting
- Module 4 · Opening the black box

Outline

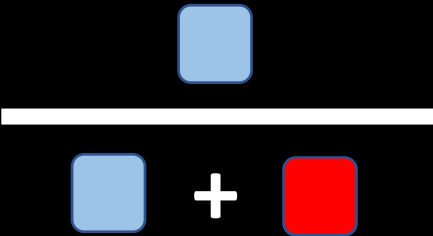
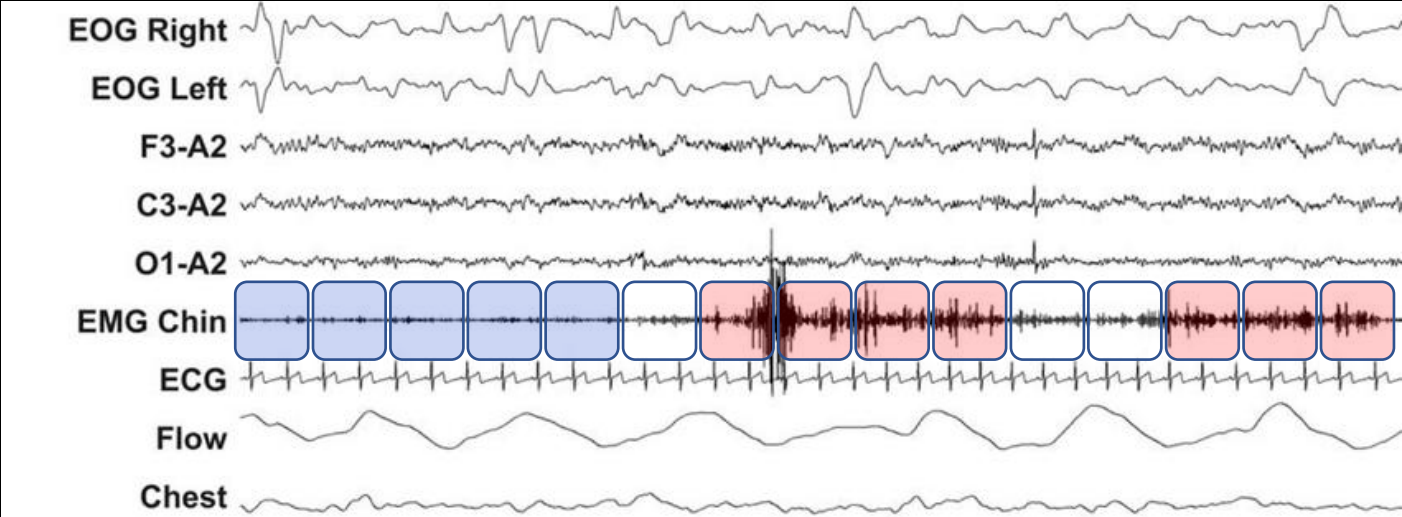
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PhysioNet CAP Sleep Database

	RBD	control
Original data	22	16
Exclude	-1 (Duplicated data; rbd10, rbd11)	-4 (Low signal sampling rate; n6, n7, n9, n12) -1 (No EMG channel; n16) -3 (Broken EMG signal; n13, n14, n15)
Final patients	21	8



REM sleep without atonia



Behav Neurol. 2010;22(3-4):91-9.

	AASM 2023	Our label	Ferri REM atonia index
Epoch duration	Tonic: 30 sec Phasic: 3 sec	3 sec	1 sec
Amplitude	Tonic: 2X NREM background Phasic: 4x NREM background	Tonic: 2 μ V Phasic: 3 μ V	≤ 1 , 1–2, $> 2\mu$ V
RBD Threshold	Not clearly defined	$> 18\%$	RAI < 0.8 $p(\leq 1\mu V) / (1 - p(1-2\mu V))$

Confusion Matrix (Type I and II Error):

		Condition (as determined by "Gold standard")		
		Condition positive	Condition negative	
Test outcome	Test outcome positive	True positive	False positive (Type I error)	Precision = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Test outcome positive}}$
	Test outcome negative	False negative (Type II error)	True negative	Negative predictive value = $\frac{\Sigma \text{ True negative}}{\Sigma \text{ Test outcome negative}}$
		Sensitivity = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Condition positive}}$	Specificity = $\frac{\Sigma \text{ True negative}}{\Sigma \text{ Condition negative}}$	Accuracy

Precision:

$$PPV = \frac{TP}{TP + FP}$$

Negative Predictive Value:

$$NPV = \frac{TN}{TN + FN}$$

Recall (Sensitivity):

$$Sensitivity = \frac{TP}{TP + FN}$$

Specificity:

$$Specificity = \frac{TN}{TN + FP}$$

F1 Score:

$$F1 \text{ Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

1 = Perfect

0 = Poor

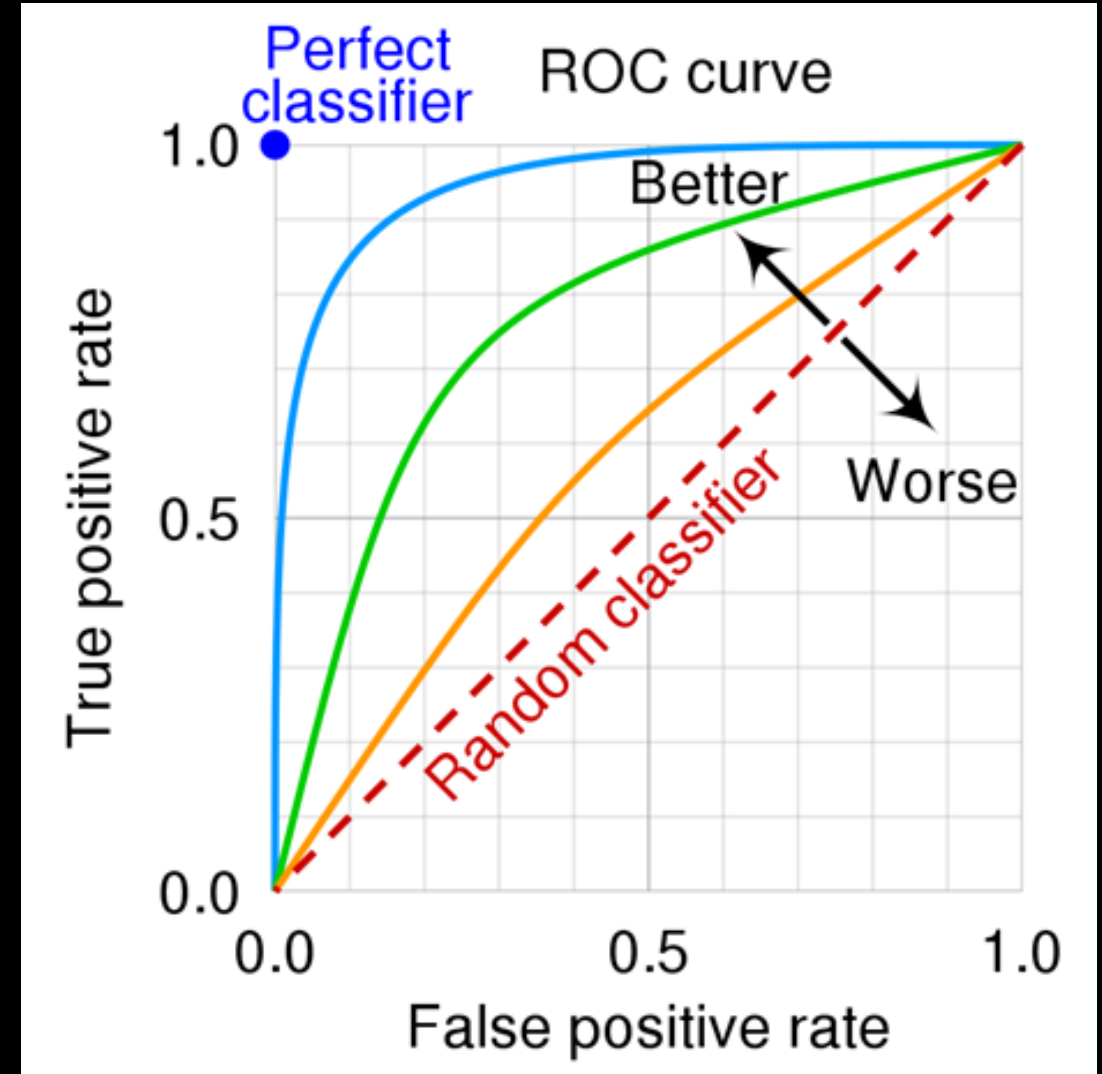
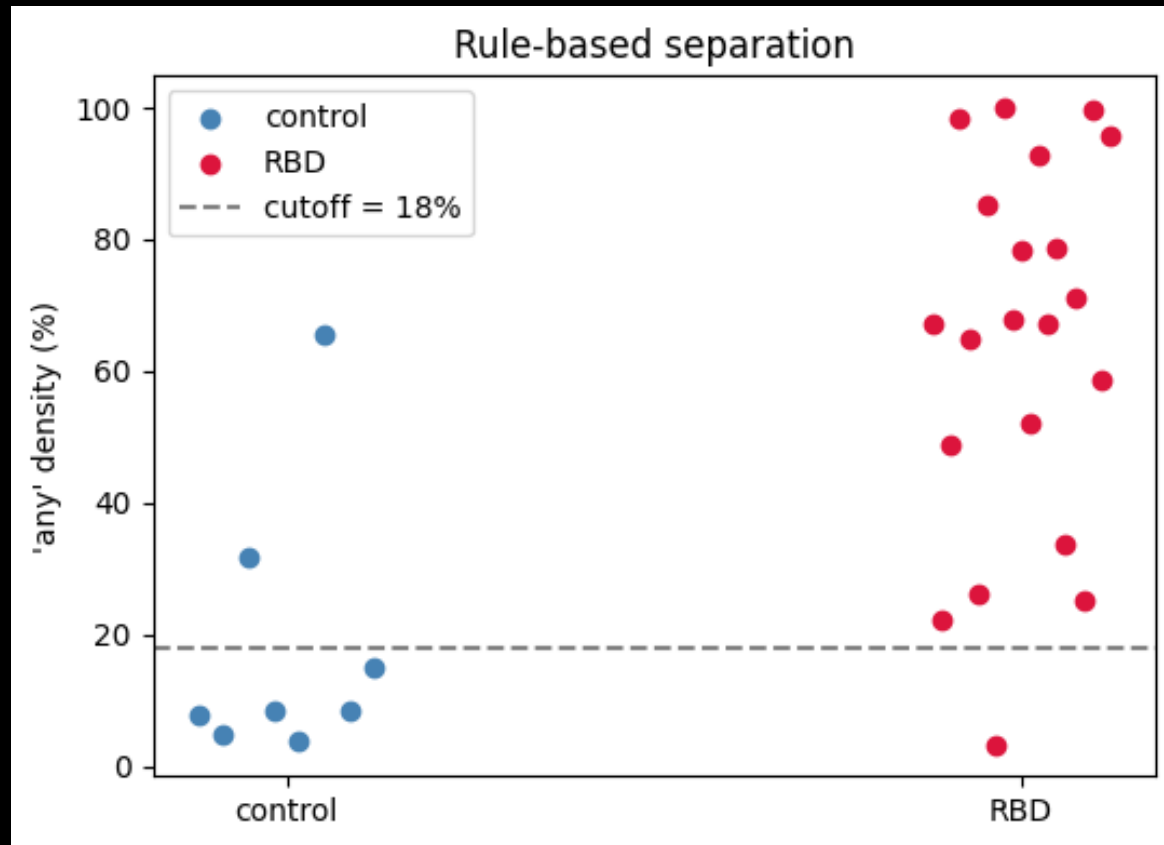
False Negative Rate:

$$FNR = \frac{FN}{FN + TP}$$

False Positive Rate:

$$FPR = \frac{FP}{FP + TN}$$

Receiver Operating Characteristic



Po-yuLin	Upload notebooks	0ab020d · now	🕒 13 Commits
data preprocessing scripts	Upload notebooks	now	
README.md	Upload notebooks	now	
setup_instructions.md	Upload notebooks	now	
workshop_notebook.ipynb	Upload notebooks	now	
workshop_notebook_answers.ipynb	Upload notebooks	now	

About

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Releases 1

📦 Workshop data v1 (RSWA)

Latest

2 weeks ago



https://github.com/Po-yuLin/AOC-IFCN_workshop



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Google Colaboratory

Suggested apps



Text Editor



Mindmap



XML, JSON Viewer



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Platform we use

The screenshot displays the Google Colab environment within the Visual Studio Code editor. The interface is divided into several sections:

- Top Bar:** Includes the Colab logo, "Welcome To Colab", a warning "Cannot save changes", and a menu with "File", "Edit", "View", "Insert", "Runtime", "Tools", and "Help". On the right, there are "Share" and "Connect" buttons.
- Table of contents (Left Panel):** Lists sections such as "Welcome to Colab!", "Google Colab is available in V...", "Access popular AI model...", "Explore the Gemini API", "Getting started", "Data science", "Machine learning", "More Resources", and "Featured examples".
- Main Editor Area:**
 - Welcome to Colab!** A section with the heading "Google Colab is available in VS Code!" and three icons (Colab, heart, VS Code). It includes instructions on how to install the Google Colab extension in VS Code.
 - Access popular AI models via Google-Colab-AI Without an API Key** A section explaining that all users have access to most popular LLMs via the `google-colab-ai` Python library.
 - Explore the Gemini API** A section with a code cell containing the following Python code:

```
from google.colab import ai
response = ai.generate_text("What is the capital of France?")
```
- Gemini Chat (Right Panel):** A chat interface with the heading "Hello, Po-Yu" and the question "How can I help you today?". It includes three suggested prompts: "How do I upload a dataset to this notebook?", "Show me how to install and import common data science libraries", and "Write a code snippet to print 'Hello, Colab!'". At the bottom, there is an "Empty cell" button and a "Gemini 2.5 Flash" model selector.

Large language model (LLM) assisted coding

- Generate from scratch: state your goal clear.
- Modify existing code: what you see, what you want to modify.
- Want to understand the code: ask LLM to explain.
- Encountering bug: paste the code, bug information (including traceback) to the LLM.

Outline

- Module 1 · From PSG signals to training data
- **Module 2 · Teaching a 1D-CNN to detect RSWA**
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A neuron (perceptron) in artificial neural network

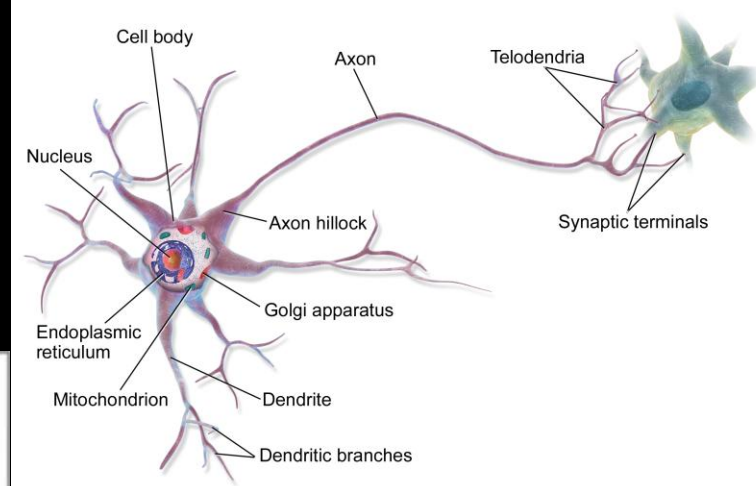
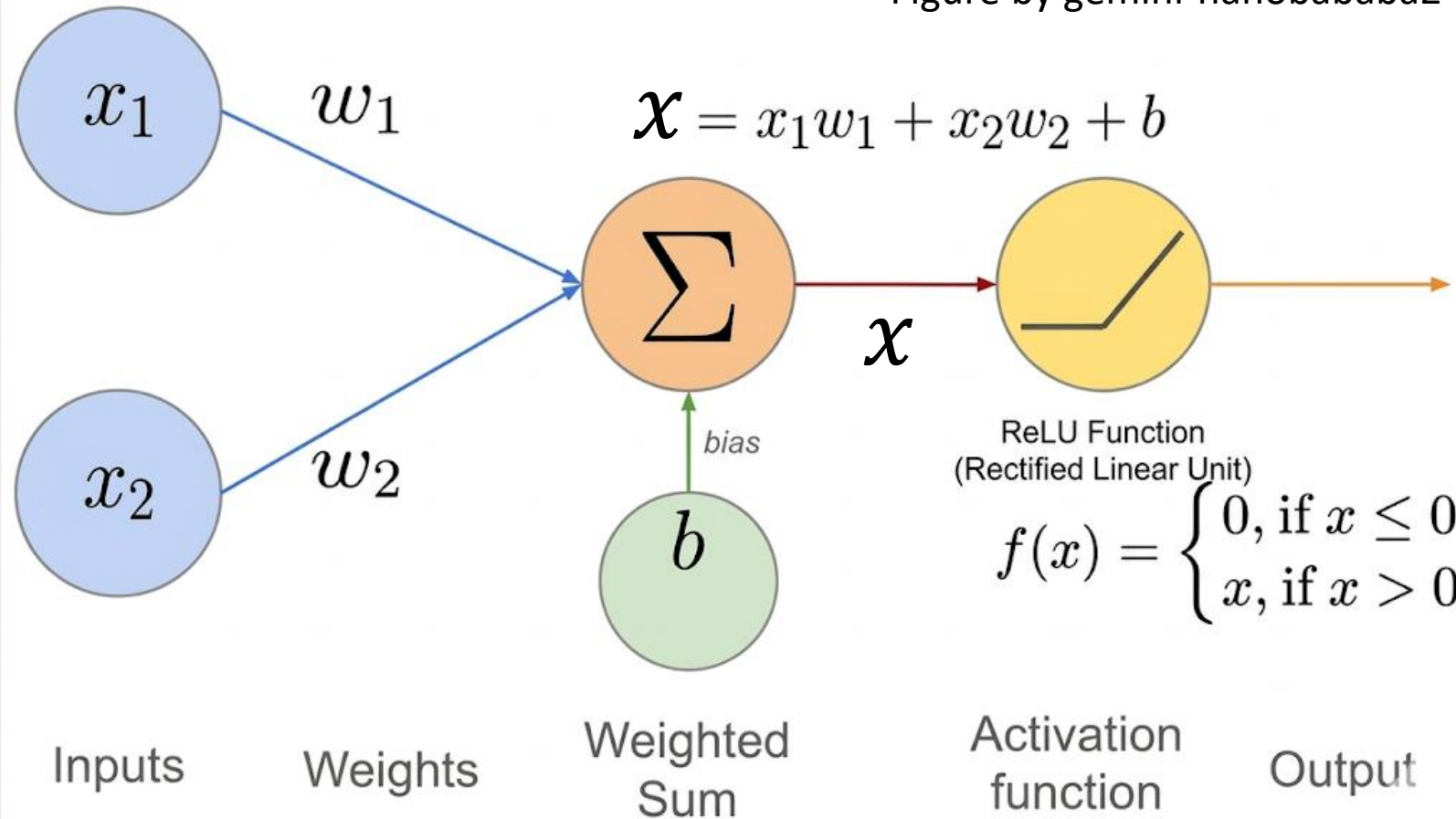


Figure by gemini-nanobababa2

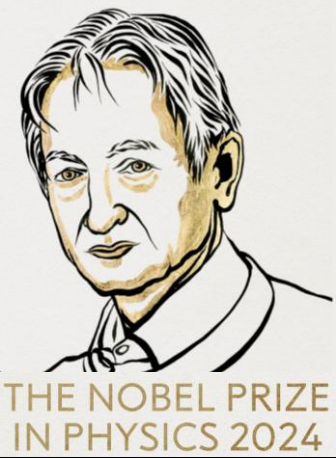
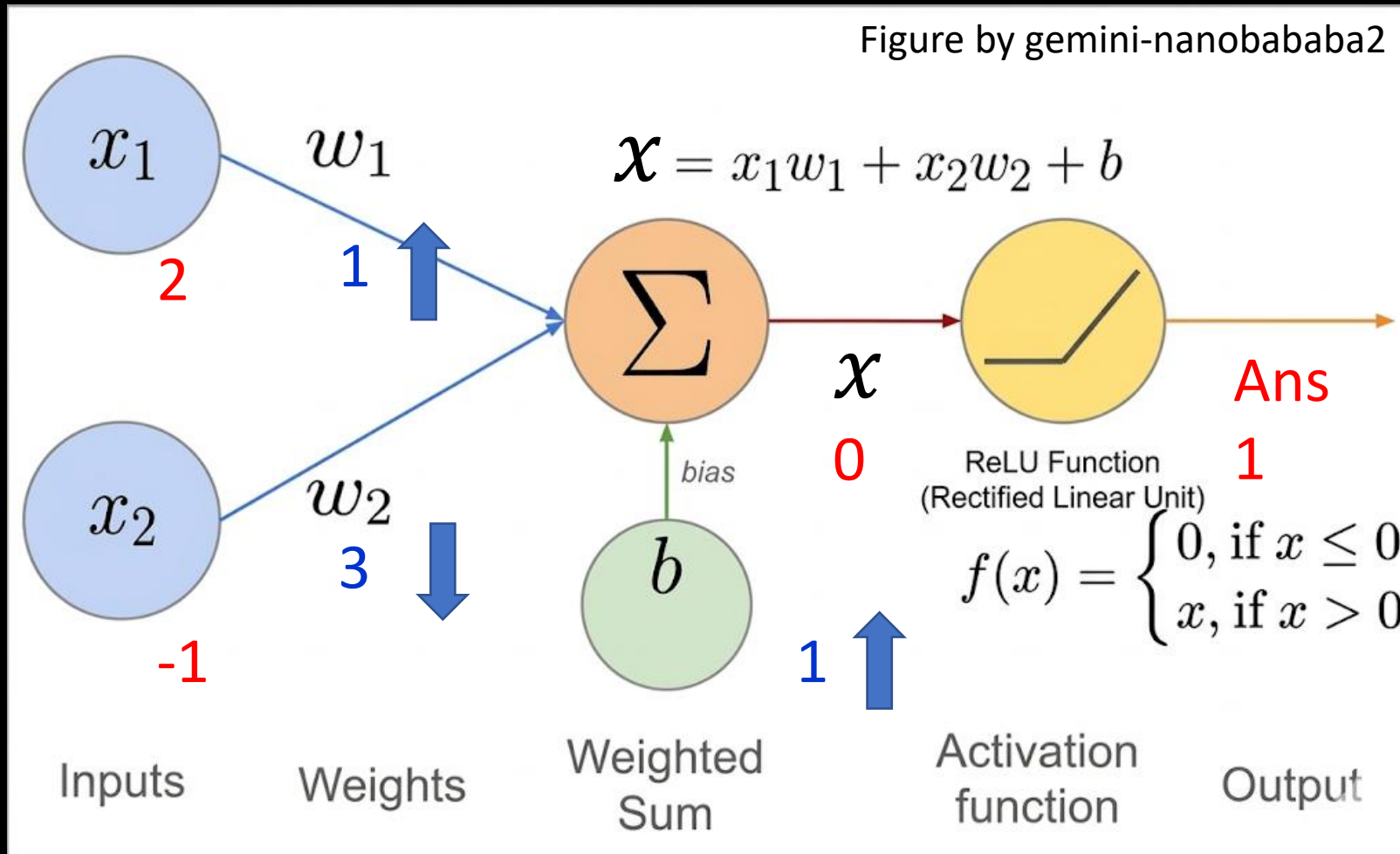


<https://en.wikipedia.org/wiki/Neuron>

- w_1 , w_2 , b are parameters
- Learning, is finding the best parameter

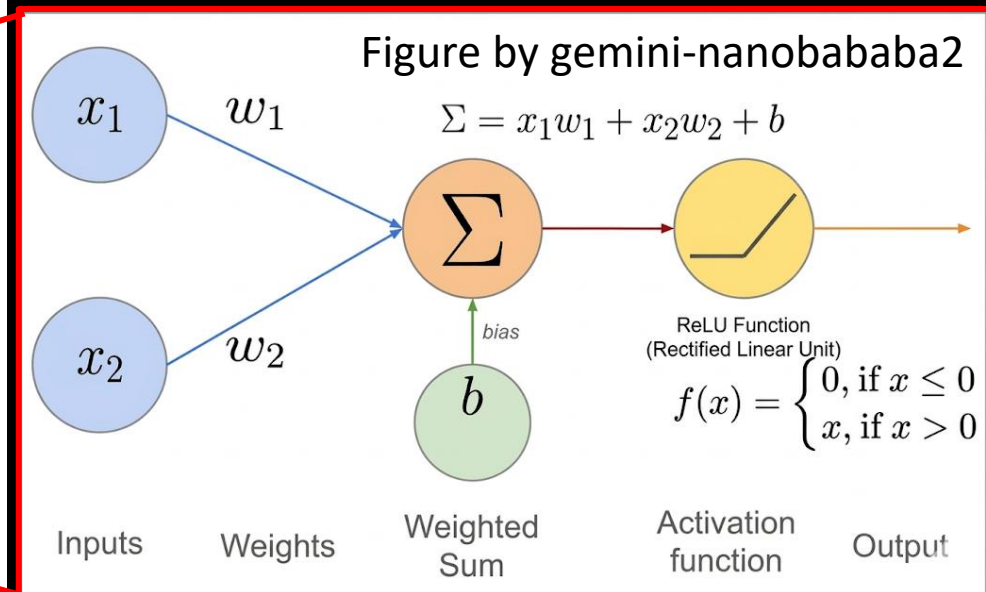
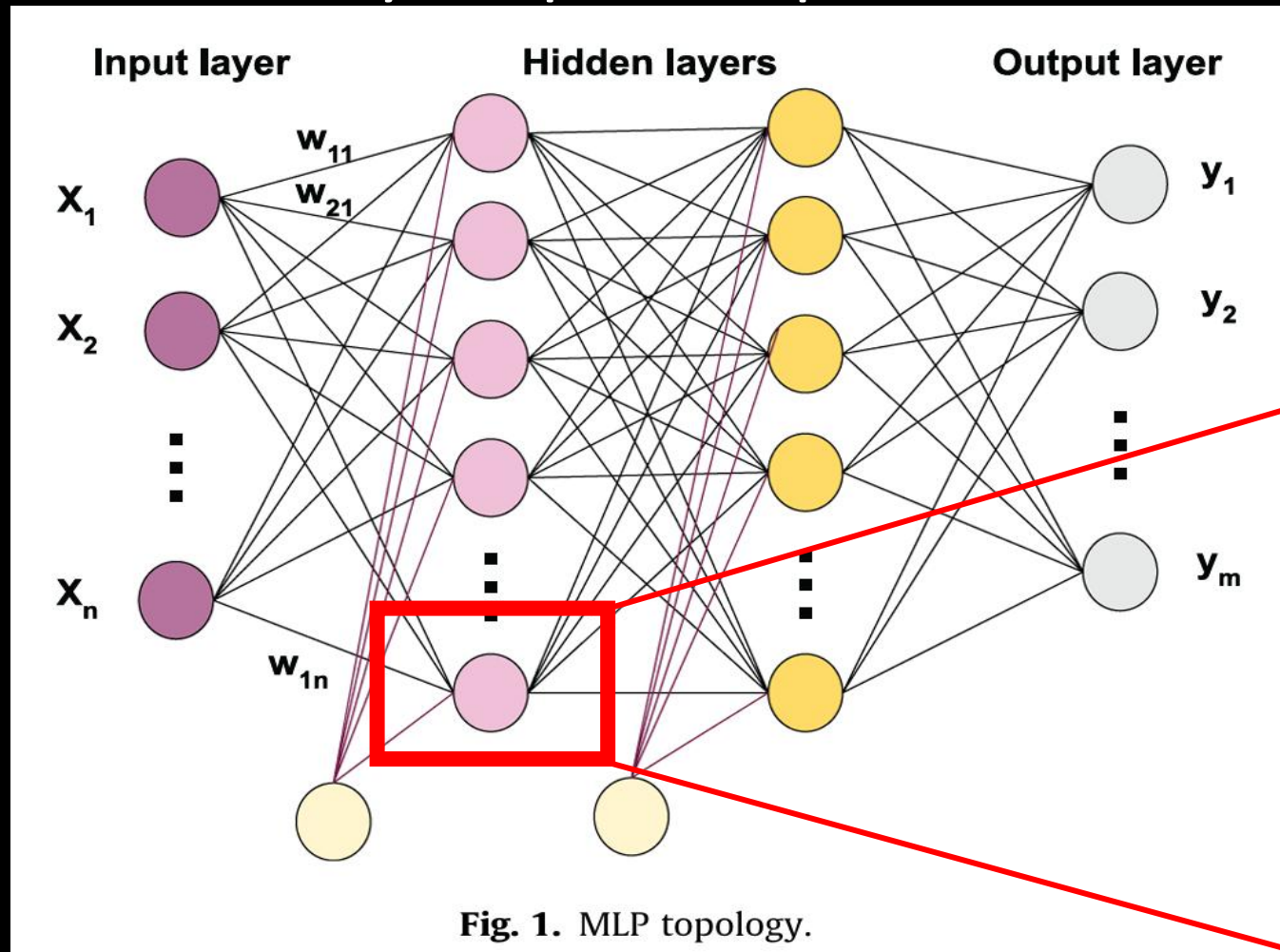
How to find better parameter?

$$\bullet \frac{-d \text{Error}}{d w_1} = \frac{-d \text{Error}}{d f(x)} \frac{d f(x)}{d x} \frac{\partial x}{\partial w_1} = -x_1$$



Back propagation

How to learn more things? multi-layer perceptron



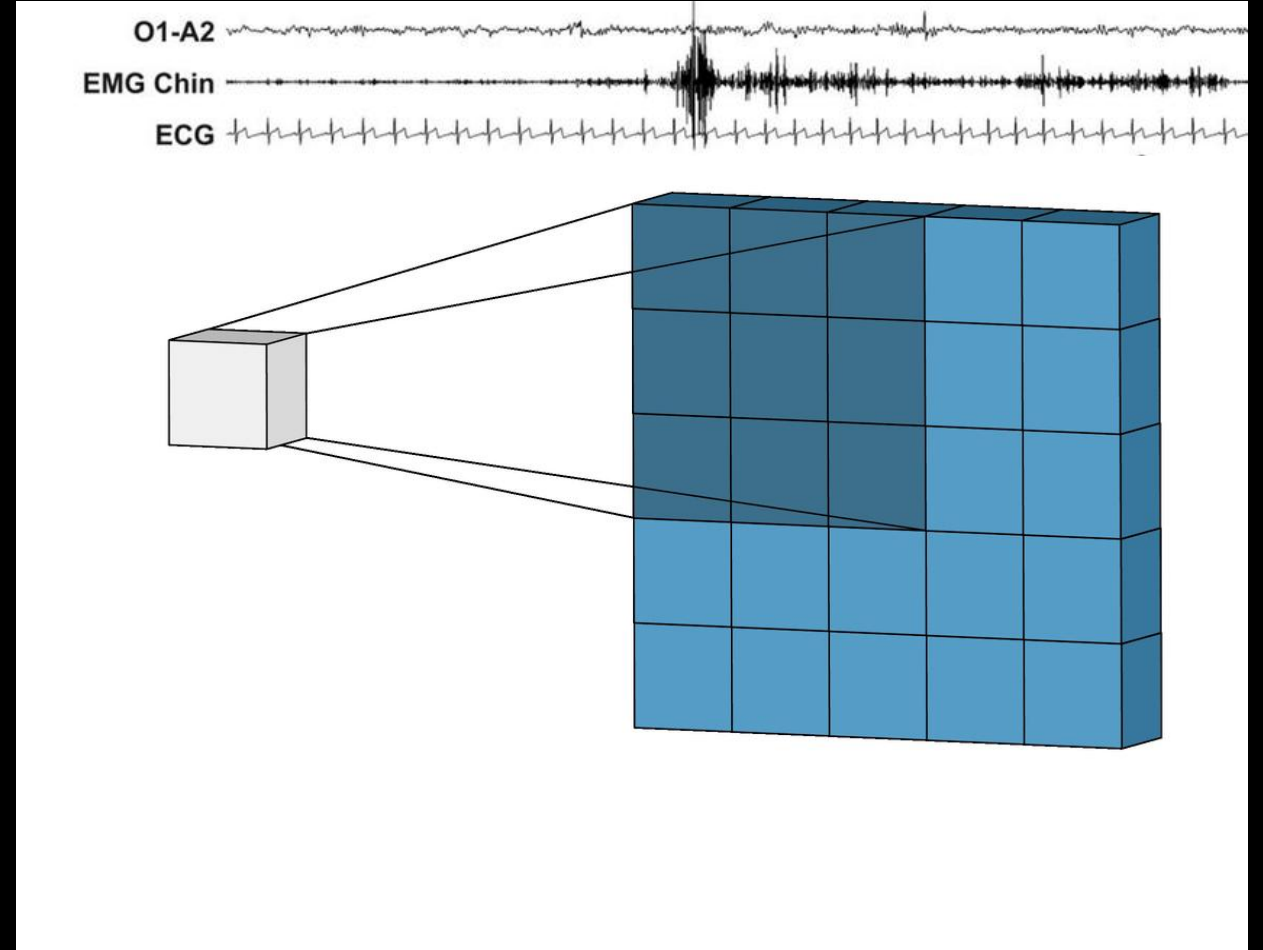
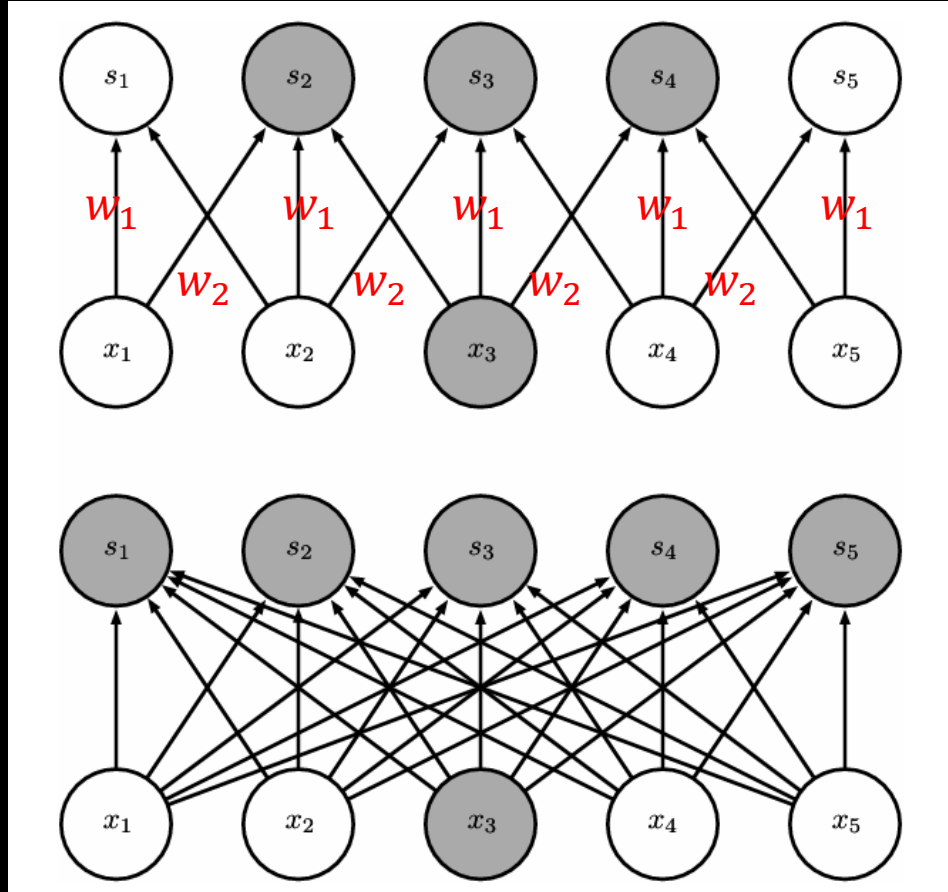
Why not make it very large?

- $x_n: (x_{n1}, x_{n2}, x_{n3}, x_{n4})$ is a patient with 4 features
- $w_1x_{11} + w_2x_{12} + w_3x_{13} + w_4x_{14} = b_1$
- $w_1x_{21} + w_2x_{22} + w_3x_{23} + w_4x_{24} = b_2$
- $w_1x_{41} + w_2x_{42} + w_3x_{43} + w_4x_{44} = b_4$

In fact, neural network is not linear.

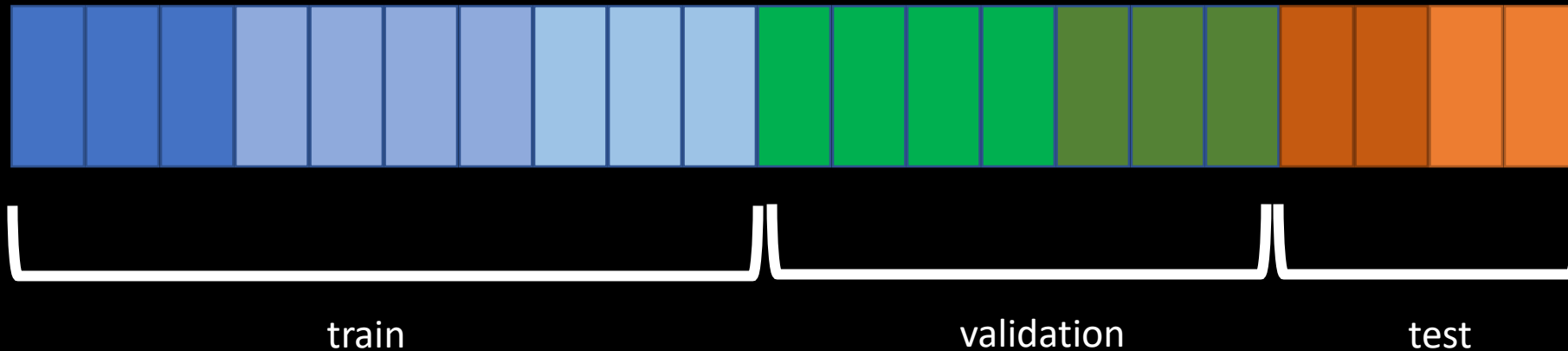
This is only for a taste and having intuition.

How to reduce parameter?



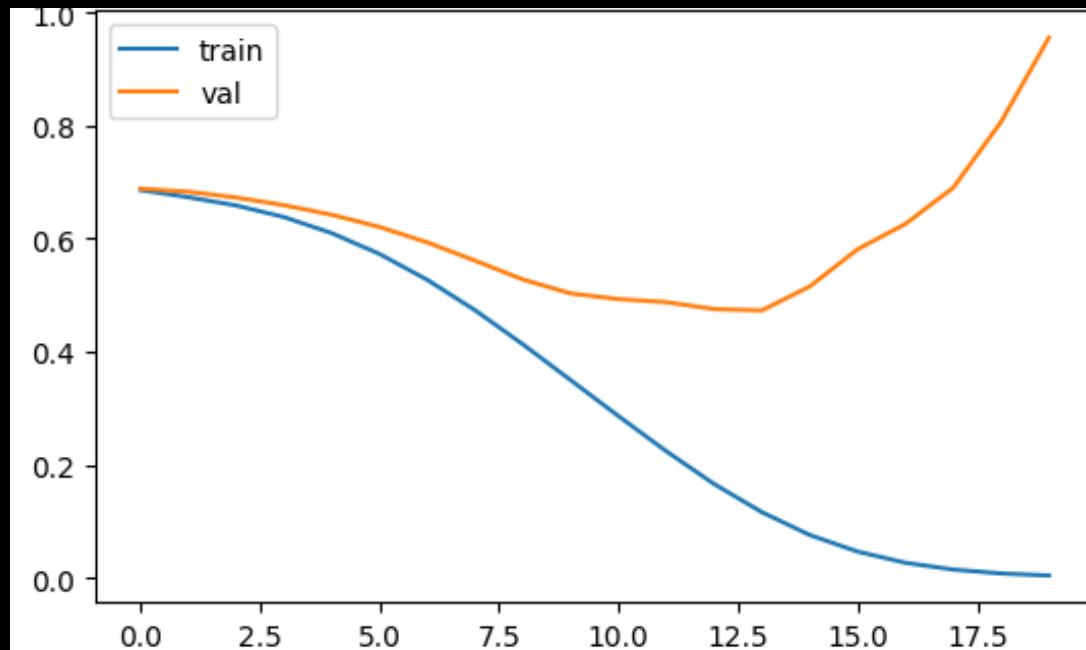
Data preparation

- Patient based separation
- Train: update parameter
- Validation: inform and adjust the training process
- Test: evaluate the final performance

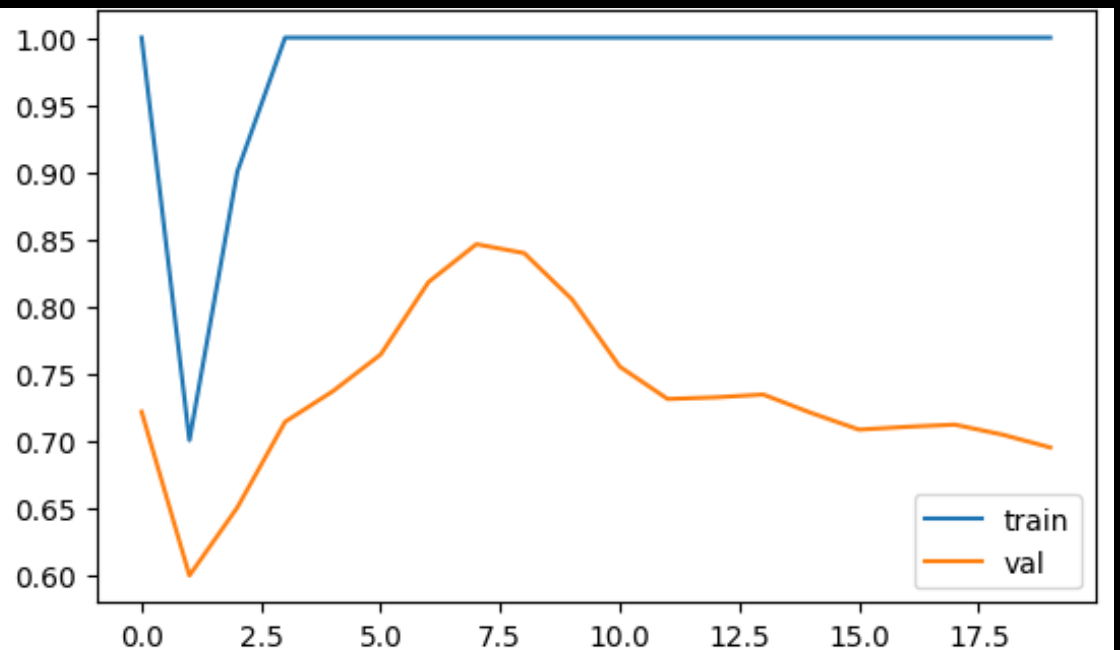


Overfit

Error



Accuracy

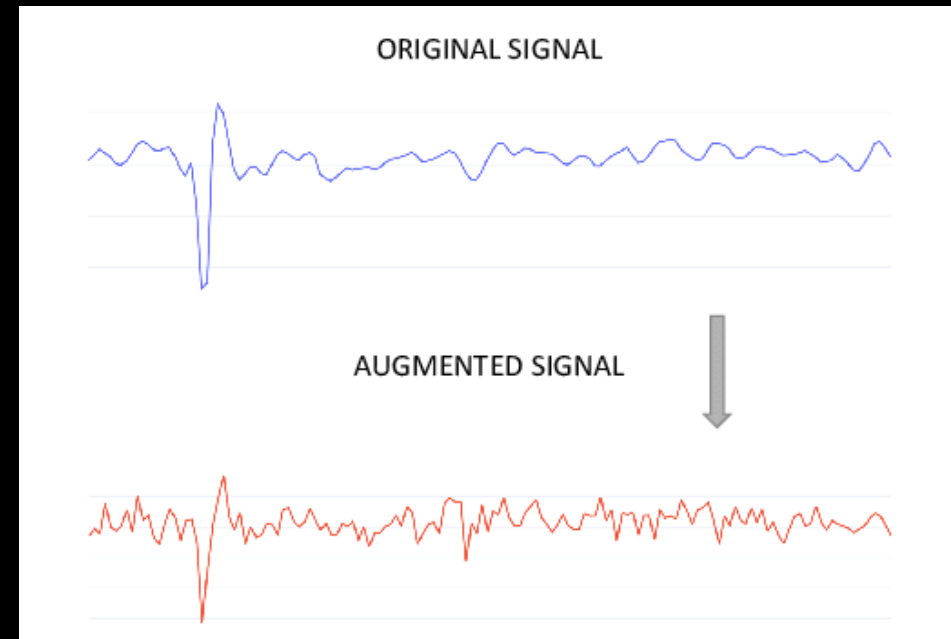


Outline

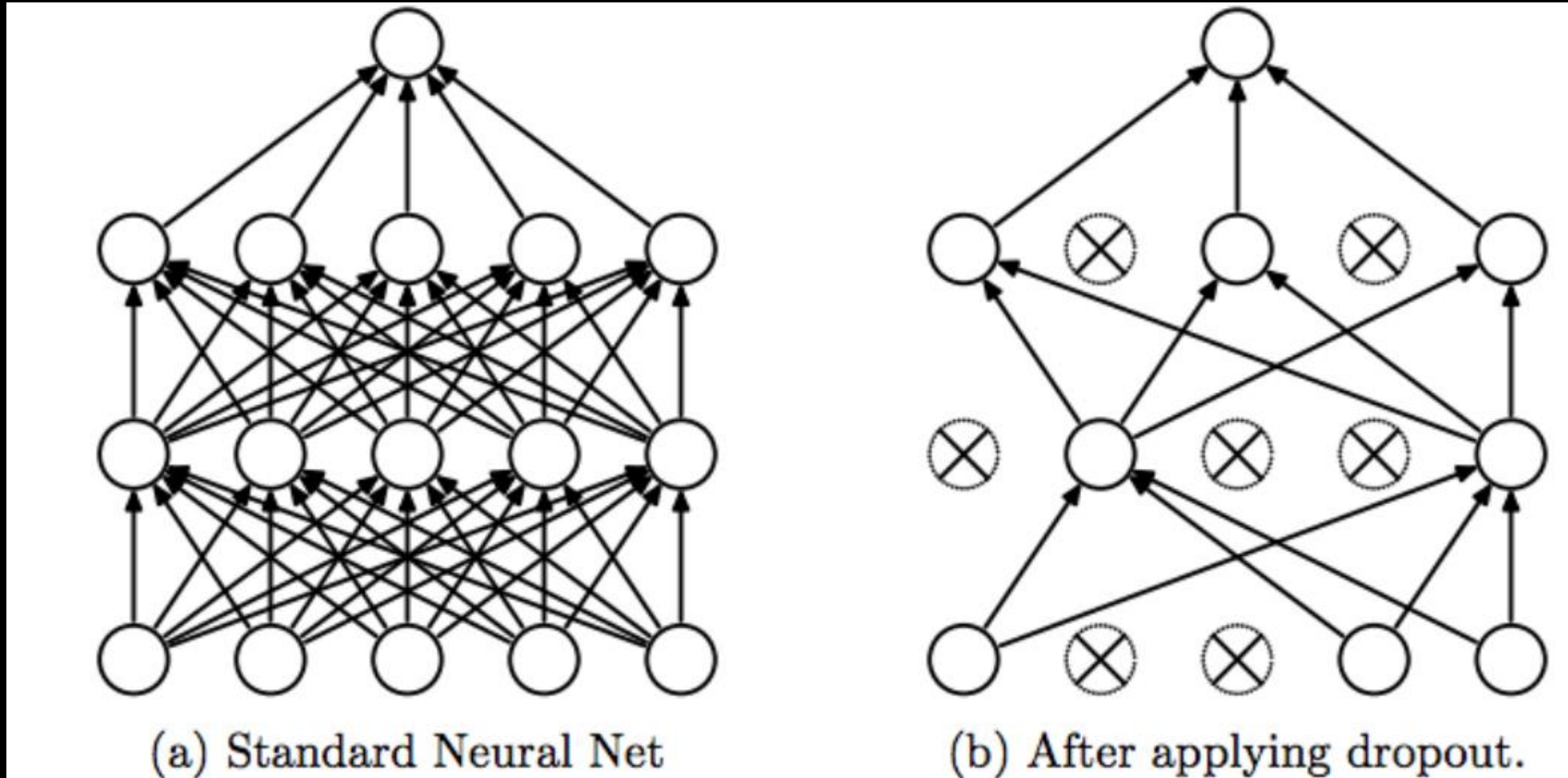
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How to prevent memorizing answers

- Fundamental solution: more questions in question bank
- Data augmentation: change question a little bit each time.
 - Add noise
 - Time shift
 - Phase angle rotation
 - ...



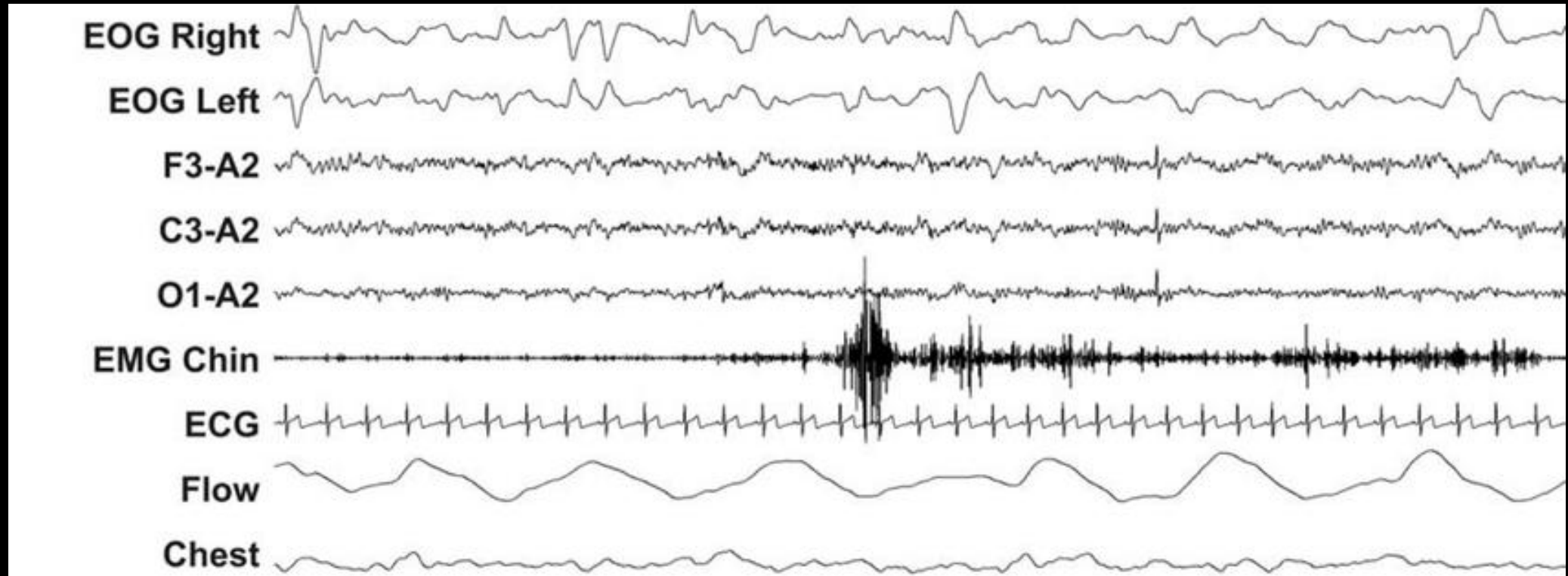
Dropout



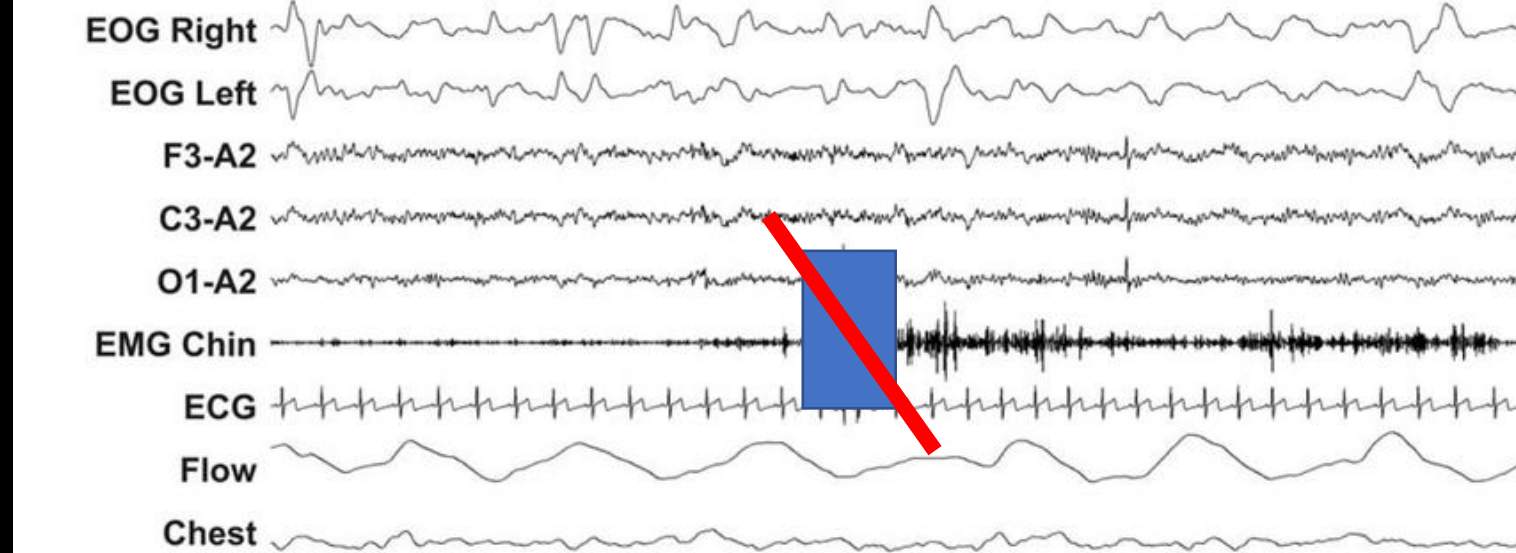
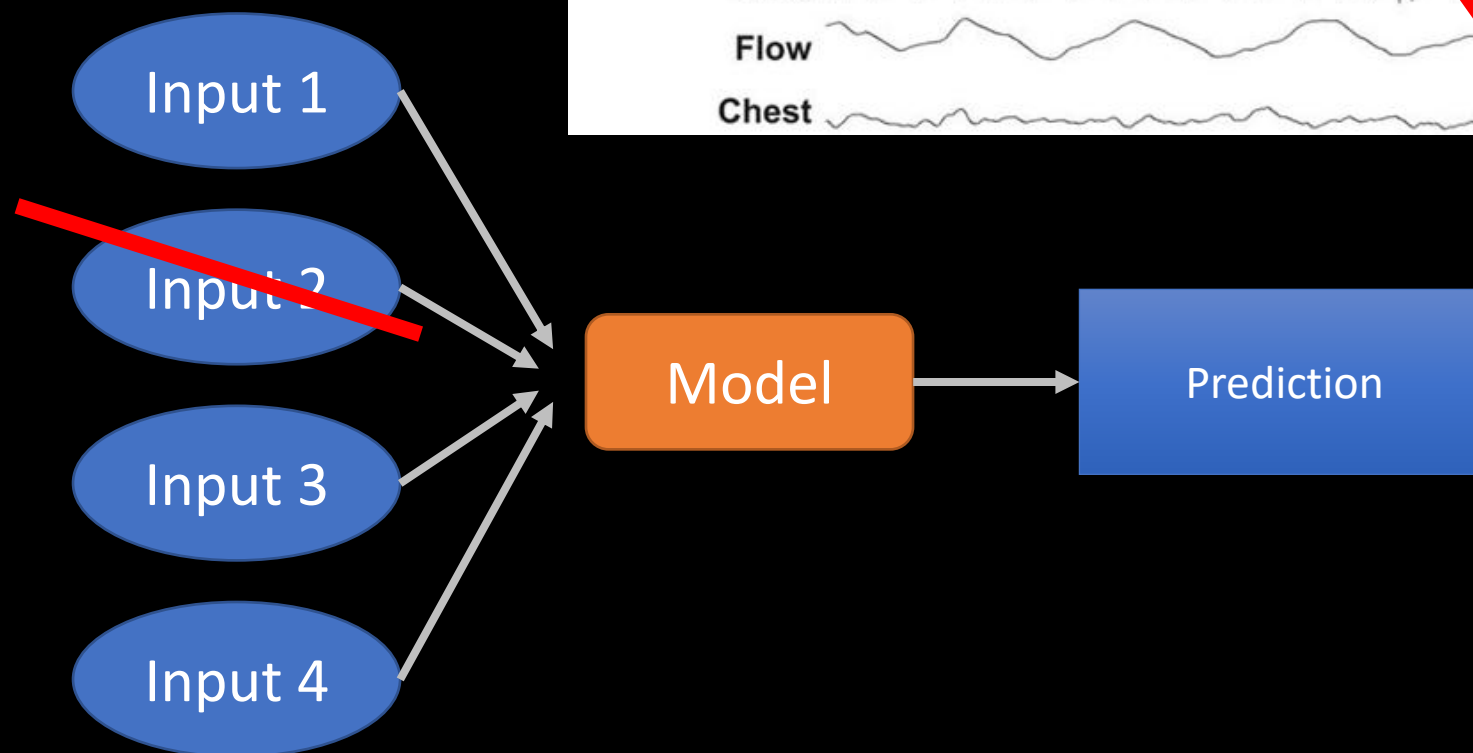
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Which part of the input affect model prediction?



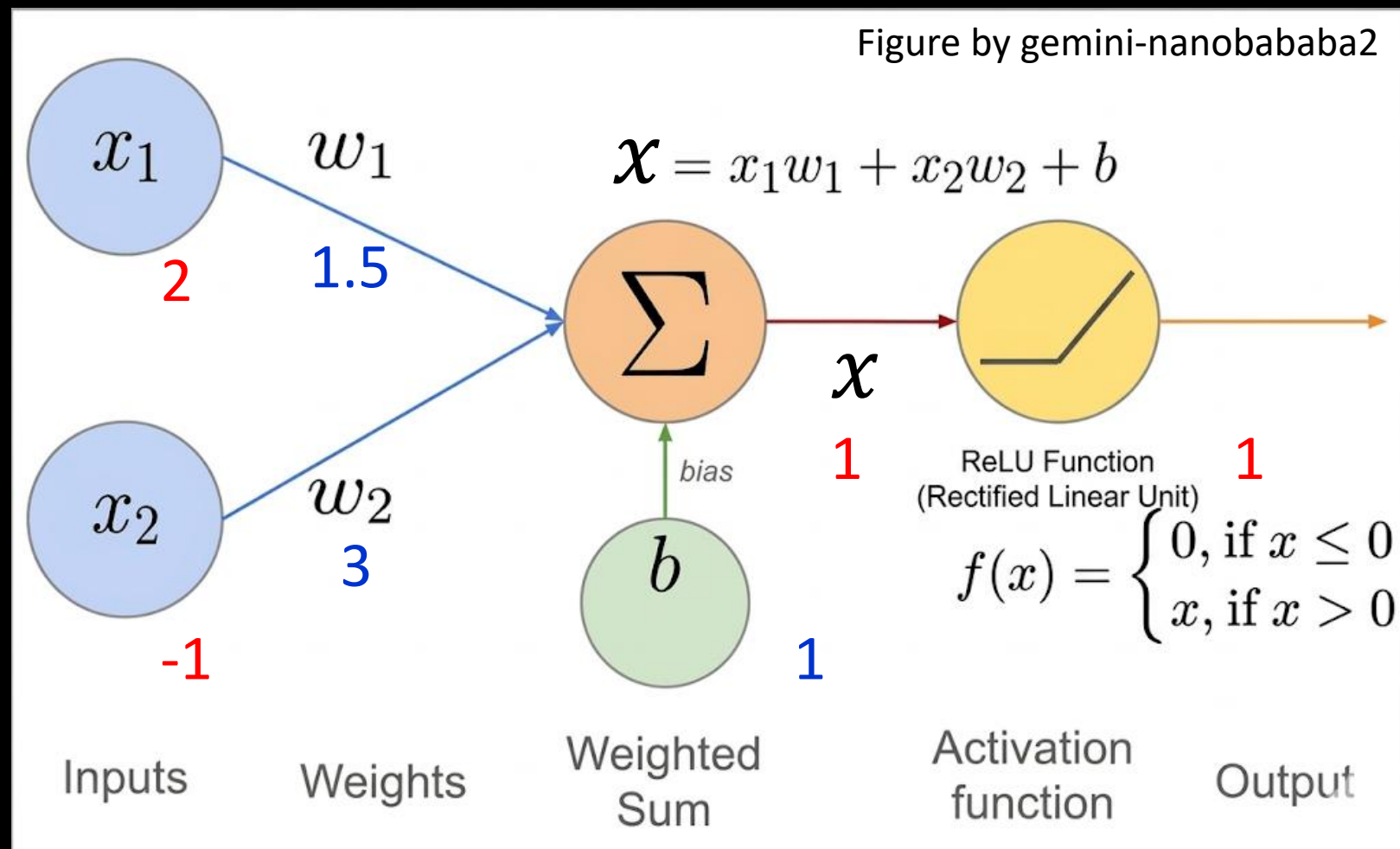
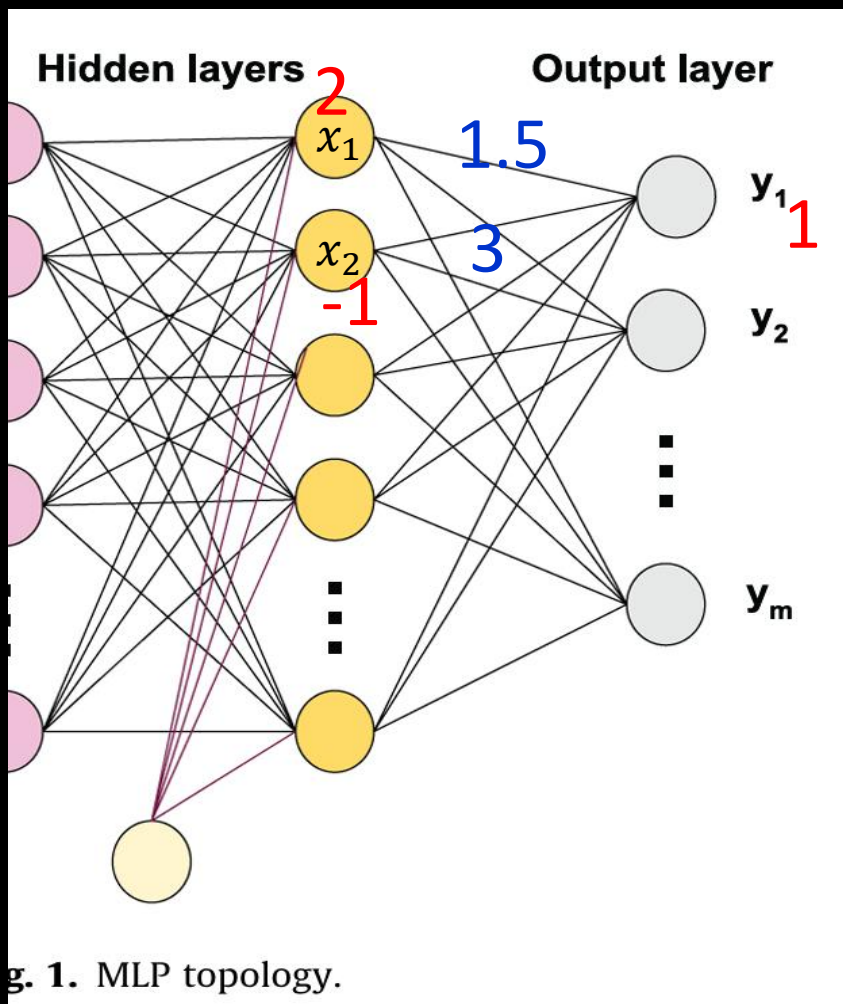
Ablation



How hard is x_1 and x_2 push the output toward current "1"?

Grad-Cam

$$Push_{x_1} = x_1 \frac{d \text{Output}}{d x_1} \quad F=ma$$



Deep learning in biomedical data

- Catch implicit relationship between data and outcome.
 - Sometimes, we only have label, but not clear cut criteria.
 - High dimension, non-linear, less feature engineering.
- Why doing it in REM sleep with atonia?
 - Sometimes, we didn't have all channels.
- The key beyond engineering is always “more high quality data”.



[https://github.com
/Po-yuLin/AOC-
IFCN_workshop](https://github.com/Po-yuLin/AOC-IFCN_workshop)



My youtube channel with
recorded lectures

Email:p88124019@gs.ncku.edu.tw

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